2.47 2nd Order DEs: Introducing Damping

Worked Solutions

1) a

$$x = 4e^{-t} + 2e^{-2t}.$$

The solution corresponds to the auxiliary equation having roots -1 and -2, which are distinct real negative roots. Therefore, the differential equation is modelling Heavy Damping (Overdamping).

1) b

$$x = e^{-3t}(4\cos t - 3\sin t).$$

The solution corresponds to the auxiliary equation having roots $-3 \pm i$, which are complex roots with negative real part. Therefore, the differential equation is modelling Light Damping (Underdamping).

1) c

$$x = 7te^{-4t} + 4e^{-4t}.$$

The solution corresponds to the auxiliary equation having a repeated real negative root, -4. Therefore, the differential equation is modelling Critical Damping.

1) d

$$x = 2e^{-6t} + 5e^{3t}.$$

The solution corresponds to the auxiliary equation having roots -6 and 3. Since one root is positive, the solution contains a growing exponential term. Therefore, the differential equation models neither Heavy Damping (Overdamping), Critical Damping, or Light Damping (Underdamping).

1) e

$$x = e^{4t}(5\cos 2t + 8\sin 2t).$$

The solution corresponds to the auxiliary equation having roots $4 \pm 2i$, which are complex roots with positive real part. As the real part is positive, the solution contains a growing exponential term. Therefore, the differential equation models neither Heavy Damping (Overdamping), Critical Damping, or Light Damping (Underdamping).

1) f

$$x = 9e^{-t} - 3te^{-t}.$$

The solution corresponds to the auxiliary equation having a repeated real negative root, -1. Therefore, the differential equation is modelling Critical Damping.

1) g

$$x = e^{-7t}\left(3\sin\frac{t}{2} - 4\cos\frac{t}{2}\right).$$

The solution corresponds to the auxiliary equation having roots $-7 \pm \frac{1}{2}i$, which are complex roots with negative real part. Therefore, the differential equation is modelling Light Damping (Underdamping).

1) h

$$x = 7e^{-6t} - 10e^{-12t}.$$

The solution corresponds to the auxiliary equation having roots -6 and -12, which are distinct real negative roots. Therefore, the differential equation is modelling Heavy Damping (Overdamping).

1) i

$$x = 8te^{t} - 3e^{t}.$$

The solution corresponds to the auxiliary equation having a repeated real positive root, 1. Since the root is positive, the solution contains two growing exponential terms. Therefore, the differential equation models neither Heavy Damping (Overdamping), Critical Damping, or Light Damping (Underdamping).

1) j

$$x = 5e^{-4t}\cos 6t.$$

The solution corresponds to the auxiliary equation having roots $-4 \pm 6i$, which are complex roots with negative real part. Therefore, the differential equation is modelling Light Damping (Underdamping).

2) a

The auxiliary equation is $m^2 + 6m + 5 = 0$:

$$m^2 + 6m + 5 = (m+1)(m+5) = 0 \implies m = -1 \text{ or } m = -5.$$

The general solution is $y = Ae^{-x} + Be^{-5x}$. This is Heavy Damping (Overdamping).

2) b

The auxiliary equation is $m^2 + 4m + 4 = 0 = (m+2)^2$, so $m = -2$. The general solution is

$$y = Axe^{-2x} + Be^{-2x} = (Ax + B)e^{-2x}.$$

This is Critical Damping.

2) c

The auxiliary equation is $m^2 + 2m + 5 = 0$:

$$(m+1)^2 - 1 + 5 = 0 \implies (m+1)^2 = -4 \implies m = -1 \pm 2i.$$

The general solution is $y = e^{-x}(A\cos 2x + B\sin 2x)$. This is Light Damping (Underdamping).

2) d

The auxiliary equation is $m^2 + 2m + 1 = 0 = (m+1)^2$, so $m = -1$. The general solution is

$$y = Axe^{-x} + Be^{-x} = (Ax + B)e^{-x}.$$

This is Critical Damping.

2) e

The auxiliary equation is $m^2 + 11m + 30 = 0 = (m+5)(m+6)$, so $m = -5$ or $m = -6$. The general solution is $y = Ae^{-5x} + Be^{-6x}$. This is Heavy Damping (Overdamping).

2) f

The auxiliary equation is $2m^2+7m+5=0=(2m+5)(m+1)$, so $m=-\frac{5}{2}$ or $m=-1$. The general solution is

$$y=Ae^{-5x/2}+Be^{-x}.$$

This is Heavy Damping (Overdamping).

2) g

The auxiliary equation is $m^2+4m+13=0$:

$$(m+2)^2-4+13=0 \implies (m+2)^2=-9 \implies m=-2\pm 3i.$$

The general solution is $y=e^{-2x}(A\cos 3x+B\sin 3x)$. This is Light Damping (Underdamping).

2) h

The auxiliary equation is $2m^2+6m+17=0$:

$$\begin{aligned} 2(m^2+3m)+17&=0,\\ 2\left((m+\tfrac{3}{2})^2-\tfrac{9}{4}\right)+17&=0,\\ 2(m+\tfrac{3}{2})^2-\tfrac{9}{2}+17&=0,\\ (m+\tfrac{3}{2})^2&=-\tfrac{25}{4}. \end{aligned}$$

Thus $m=-\frac{3}{2}\pm\frac{5}{2}i$ and the general solution is

$$y=e^{-3x/2}\left(A\cos\frac{5}{2}x+B\sin\frac{5}{2}x\right).$$

This is Light Damping (Underdamping).

2) i

The auxiliary equation is $9m^2+30m+25=0=(3m+5)^2$, so $m=-\frac{5}{3}$. The general solution is

$$y=Axe^{-5x/3}+Be^{-5x/3}=(Ax+B)e^{-5x/3}.$$

This is Critical Damping.

3) a

This is Heavy Damping (Overdamping), where $k^2>4\omega^2$. The auxiliary equation has two distinct real roots, both of which are negative.

3) b

This is Light Damping (Underdamping), where $k^2<4\omega^2$. The auxiliary equation has two complex conjugate roots, where the real part is negative.

4) a

$$y=3e^{-x}-2e^{-2x}, \qquad \frac{dy}{dx}=-3e^{-x}+4e^{-2x}.$$

Stationary points exist when $\frac{dy}{dx}=0$:

$$-3e^{-x}+4e^{-2x}=0 \implies 4e^{-2x}=3e^{-x} \implies \frac{4}{3}=e^{x} \implies x=\ln\frac{4}{3}.$$

$$y\big|_{x=\ln(4/3)}=\frac{9}{8}.$$

So there is a stationary point at $\left(\ln\frac{4}{3},\frac{9}{8}\right)$.

4) b

$$y = 4e^{-3x} + 3e^{-x}, \qquad \frac{dy}{dx} = -12e^{-3x} - 3e^{-x}.$$

Stationary points exist when $\frac{dy}{dx} = 0$:

$$-12e^{-3x} - 3e^{-x} = 0 \implies -4e^{-3x} = e^{-x} \implies -4 = e^{2x}.$$

This has no real solutions, so there are no stationary points.

4) c

$$y = -3e^{-2x} - 6e^{-4x}, \qquad \frac{dy}{dx} = 6e^{-2x} + 24e^{-4x}.$$

Stationary points exist when $\frac{dy}{dx} = 0$:

$$6e^{-2x} + 24e^{-4x} = 0 \implies e^{-2x} = -4e^{-4x} \implies e^{2x} = -4.$$

This has no real solutions, so there are no stationary points.

4) d

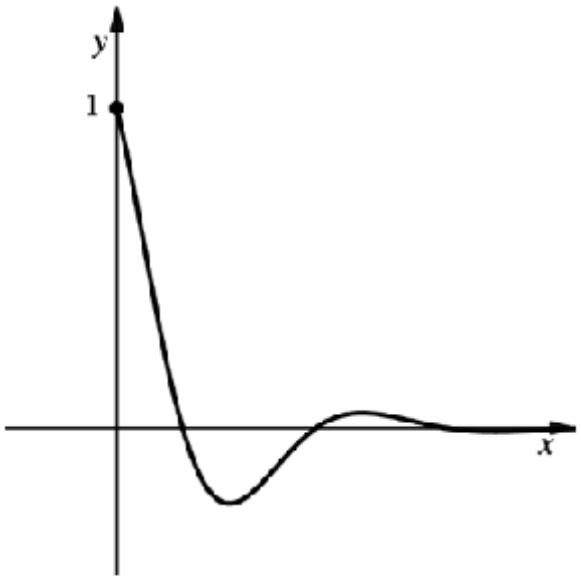
$$y = (2x-1)e^{-2x}, \qquad \frac{dy}{dx} = -2(2x-1)e^{-2x} + 2e^{-2x}.$$

Stationary points exist when $\frac{dy}{dx} = 0$:

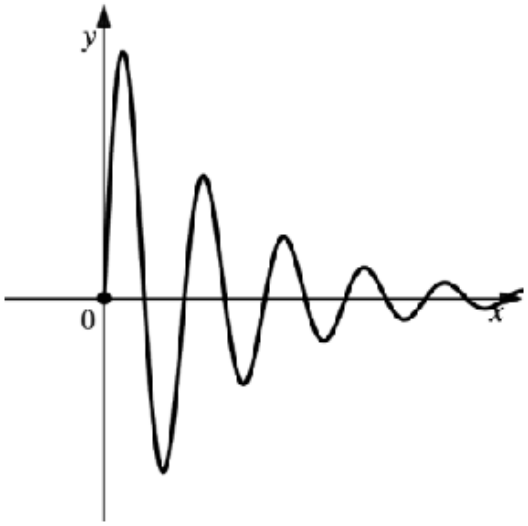
$$\begin{aligned} -2(2x-1)e^{-2x} + 2e^{-2x} &= 0, \\ (-4x+2)e^{-2x} + 2e^{-2x} &= 0, \\ (-4x+2+2)e^{-2x} &= 0, \\ (4-4x)e^{-2x} &= 0, \\ x &= 1. \end{aligned}$$

$y|_{x=1} = e^{-2}$, so there is a stationary point at $(1, e^{-2})$.

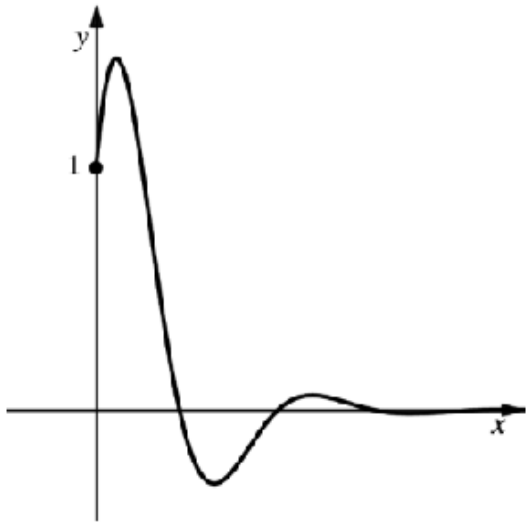
5) a



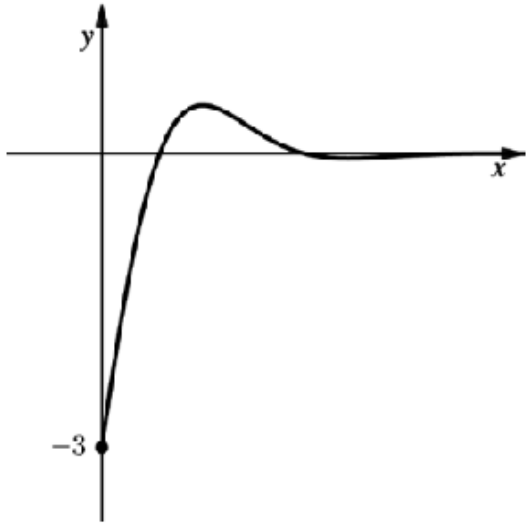
5) b



5) c



5) d



6) a

$$2\ddot{x} + 5\dot{x} + 3x = 0.$$

The auxiliary equation is $2m^2+5m+3 = 0 = (2m+3)(m+1)$, so $m = -\frac{3}{2}$ or $m = -1$. The general solution is $x = Ae^{-3t/2} + Be^{-t}$. Using $x = 2$ at $t = 0$ gives $A + B = 2$. Also $\dot{x} = -\frac{3}{2}Ae^{-3t/2} - Be^{-t}$. Using $\dot{x} = 1$ at $t = 0$ gives $-\frac{3}{2}A - B = 1$. Solving these equations gives $A = -6$ and $B = 8$. Thus

$$x = -6e^{-3t/2} + 8e^{-t}.$$

6) b

This is Heavy Damping (Overdamping) as the auxiliary equation has two distinct real roots.

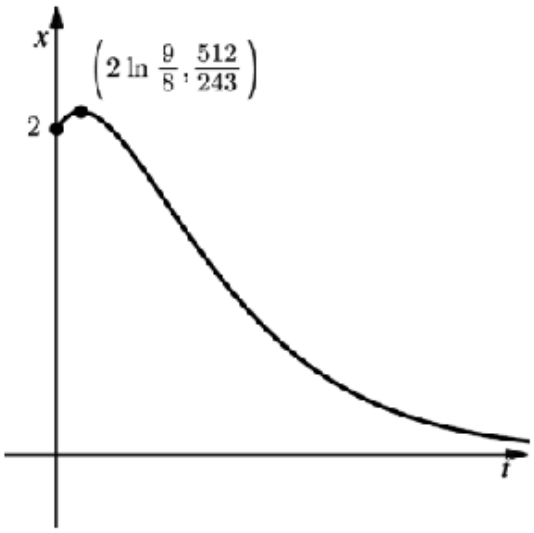
6) c

$$x = -6e^{-3t/2} + 8e^{-t}, \qquad \dot{x} = 9e^{-3t/2} - 8e^{-t}.$$

Stationary points exist when $\dot{x} = 0$:

$$0 = 9e^{-3t/2} - 8e^{-t} \implies 8e^{-t} = 9e^{-3t/2} \implies e^{t/2} = \frac{9}{8} \implies t = 2\ln\frac{9}{8}.$$

6) d



7) a

The auxiliary equation is $4m^2 + 4m + 5 = 0$:

$$\begin{aligned} 4(m^2 + m) + 5 &= 0, \\ 4\left((m + \tfrac{1}{2})^2 - \tfrac{1}{4}\right) + 5 &= 0, \\ 4(m + \tfrac{1}{2})^2 - 1 + 5 &= 0, \\ (m + \tfrac{1}{2})^2 &= -1. \end{aligned}$$

Thus $m = -\frac{1}{2} \pm i$ and $x = e^{-t/2}(A\cos t + B\sin t)$. Using $x = 2$ and $t = 0$ gives $A = 2$, so $x = e^{-t/2}(2\cos t + B\sin t)$. Differentiating,

$$\dot{x} = e^{-t/2}(-2\sin t + B\cos t) - \tfrac{1}{2}e^{-t/2}(2\cos t + B\sin t).$$

Using $\dot{x} = -1$ and $t = 0$:

$$-1 = (0 + B) - \tfrac{1}{2}(2 + 0) = B - 1,$$

so $B = 0$. The particular solution is

$$x = 2e^{-t/2}\cos t.$$

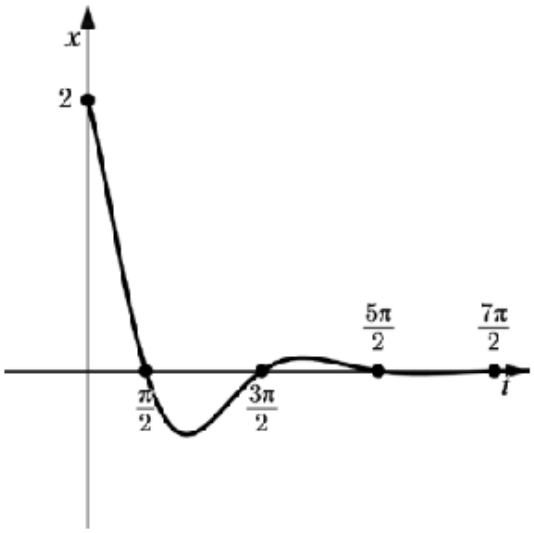
7) b

This is Light Damping (Underdamping) as the auxiliary equation has complex conjugate solutions.

7) c

$\cos t$ has period 2π, so the period of the oscillations is 2π.

7) d



8) a

$$y = 4e^{-3x} - 2e^{-2x} + 1, \qquad \frac{dy}{dx} = -12e^{-3x} + 4e^{-2x}.$$

Stationary points exist when $\frac{dy}{dx} = 0$:

$$-12e^{-3x} + 4e^{-2x} = 0 \implies e^{-2x} = 3e^{-3x} \implies e^{x} = 3 \implies x = \ln 3.$$

$y|_{x=\ln 3} = \frac{25}{27}$, so there is a stationary point at $\left(\ln 3, \frac{25}{27}\right)$.

8) b

$$y = e^{-2x} - 2e^{-3x} - 2, \qquad \frac{dy}{dx} = -2e^{-2x} + 6e^{-3x}.$$

Stationary points exist when $\frac{dy}{dx} = 0$:

$$-2e^{-2x} + 6e^{-3x} = 0 \implies 3e^{-3x} = e^{-2x} \implies 3 = e^{x} \implies x = \ln 3.$$

$y|_{x=\ln 3} = -\frac{53}{27}$, so there is a stationary point at $\left(\ln 3, -\frac{53}{27}\right)$.

8) c

$$y = 3e^{-x} - 3e^{-2x} - 1, \qquad \frac{dy}{dx} = -3e^{-x} + 6e^{-2x}.$$

Stationary points exist when $\frac{dy}{dx} = 0$:

$$-3e^{-x} + 6e^{-2x} = 0 \implies 2e^{-2x} = e^{-x} \implies 2 = e^{x} \implies x = \ln 2.$$

$y|_{x=\ln 2} = -\frac{1}{4}$, so there is a stationary point at $\left(\ln 2, -\frac{1}{4}\right)$.

8) d

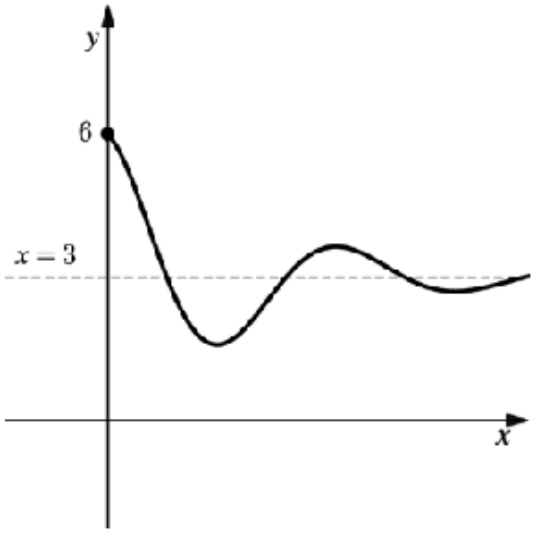
$$y = (3 - 9x)e^{-3x} + 2, \qquad \frac{dy}{dx} = -3(3 - 9x)e^{-3x} - 9e^{-3x}.$$

Stationary points exist when $\frac{dy}{dx} = 0$:

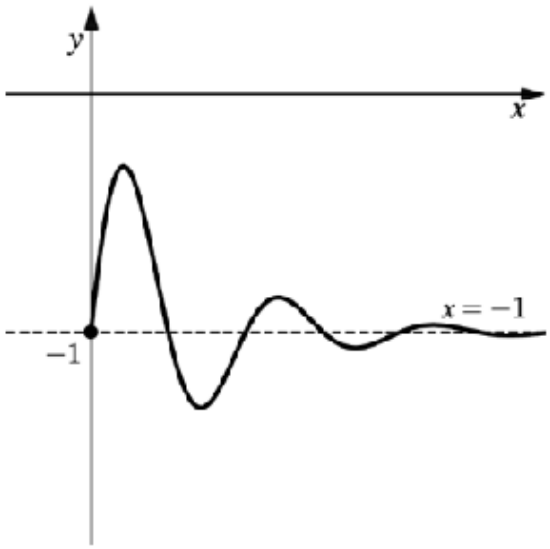
$$\begin{aligned} -3(3-9x)e^{-3x} - 9e^{-3x} &= 0, \\ (3-9x)e^{-3x} + 3e^{-3x} &= 0, \\ (3-9x+3)e^{-3x} &= 0, \\ (6-9x)e^{-3x} &= 0, \\ x &= \frac{2}{3}. \end{aligned}$$

$y|_{x=2/3} = -3e^{-2} + 2$, so there is a stationary point at $\left(\frac{2}{3}, -3e^{-2} + 2\right)$.

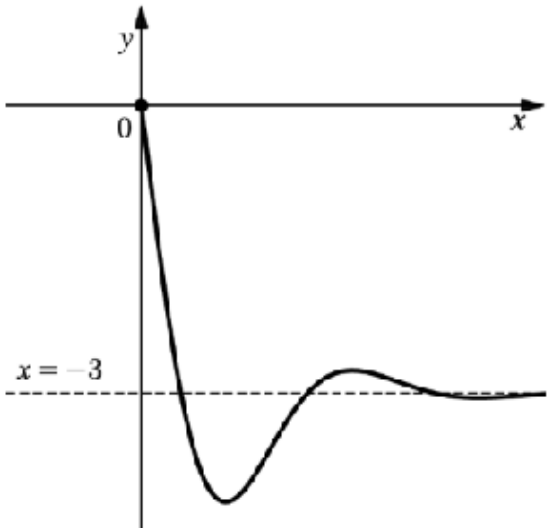
9) a



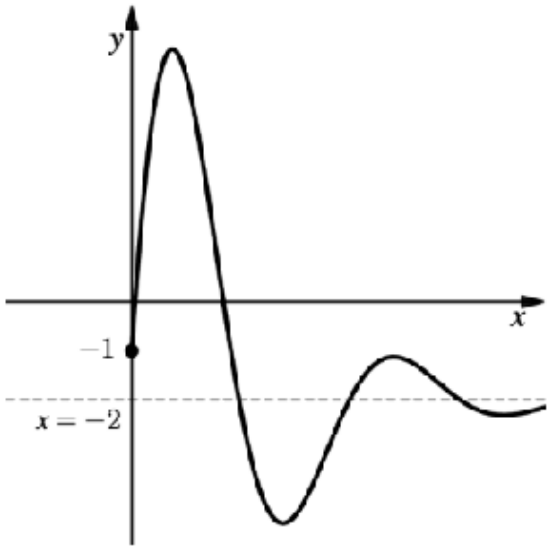
9) b



9) c



9) d



10) a

$$\ddot{x} + 6\dot{x} + 8x = 8.$$

The homogeneous equation is $\ddot{x} + 6\dot{x} + 8x = 0$. The auxiliary equation is $m^2 + 6m + 8 = 0 = (m+2)(m+4)$, so $m = -2$ or $m = -4$. The complementary function is $x = Ae^{-2t} + Be^{-4t}$. Try a constant particular integral $x = k$, so $\dot{x} = 0$ and $\ddot{x} = 0$. Substitution gives $8k = 8$, so $k = 1$. Thus $x = Ae^{-2t} + Be^{-4t} + 1$. Using $x = 2$ and $t = 0$ gives $A + B = 1$. Also $\dot{x} = -2Ae^{-2t} - 4Be^{-4t}$. Using $\dot{x} = 2$ and $t = 0$ gives $-2A - 4B = 2$. Solving gives $A = 3$ and $B = -2$. The particular solution is

$$x = 3e^{-2t} - 2e^{-4t} + 1.$$

10) b

This is Heavy Damping (Overdamping) as the auxiliary equation has two distinct real roots.

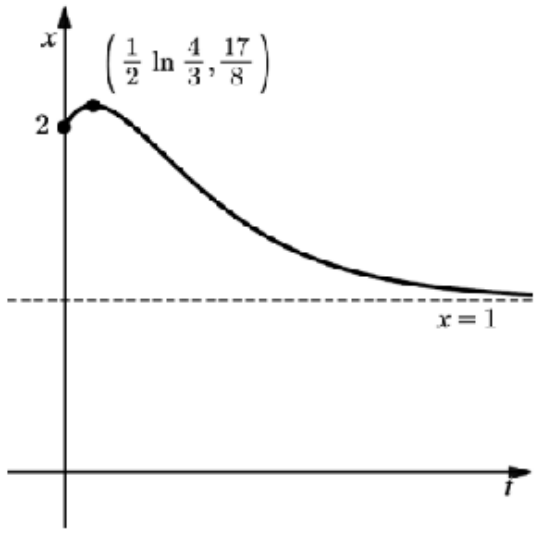
10) c

$$x = 3e^{-2t} - 2e^{-4t} + 1, \qquad \dot{x} = -6e^{-2t} + 8e^{-4t}.$$

Stationary points exist when $\dot{x} = 0$:

$$0 = -6e^{-2t} + 8e^{-4t} \implies 3e^{-2t} = 4e^{-4t} \implies e^{2t} = \frac{4}{3} \implies t = \frac{1}{2}\ln\frac{4}{3}.$$

10) d



11) a

$$16\ddot{x} + 8\dot{x} + x = -3.$$

The homogeneous equation is $16\ddot{x} + 8\dot{x} + x = 0$. The auxiliary equation is $16m^2 + 8m + 1 = 0 = (4m+1)^2$, so $m = -\frac{1}{4}$. The complementary function is $x = (At+B)e^{-t/4}$. Try a constant particular integral $x = k$, so $\dot{x} = 0$ and $\ddot{x} = 0$. Substitution gives $k = -3$. Thus $x = (At + B)e^{-t/4} - 3$. Using $x = 1$ and $t = 0$ gives $1 = B - 3$, so $B = 4$. Then

$$x = (At + 4)e^{-t/4} - 3, \qquad \dot{x} = -\tfrac{1}{4}(At + 4)e^{-t/4} + Ae^{-t/4}.$$

Using $\dot{x} = 1$ and $t = 0$ gives $1 = -1 + A$, so $A = 2$. The particular solution is

$$x = (2t + 4)e^{-t/4} - 3.$$

11) b

This is Critical Damping as the auxiliary equation has two repeated real roots.

11) c

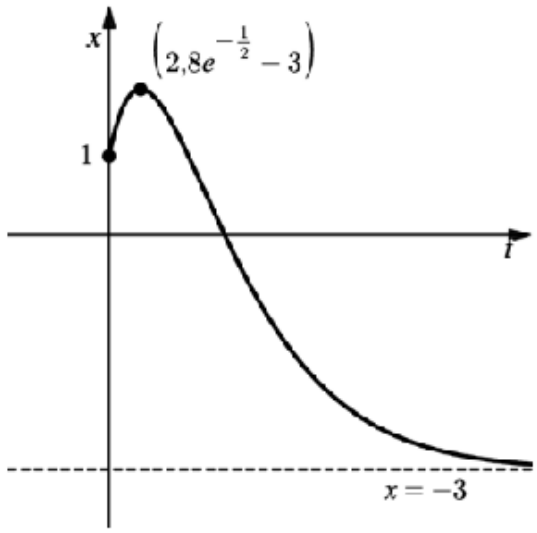
$$x = (2t + 4)e^{-t/4} - 3, \qquad \dot{x} = -\tfrac{1}{4}(2t + 4)e^{-t/4} + 2e^{-t/4}.$$

Stationary points exist when $\dot{x} = 0$:

$$\begin{aligned} 0 &= -\tfrac{1}{4}(2t+4)e^{-t/4} + 2e^{-t/4}, \\ 0 &= -\tfrac{1}{4}(t+2)e^{-t/4} + e^{-t/4}, \\ 0 &= (t+2)e^{-t/4} - 4e^{-t/4}, \\ 0 &= (t+2-4)e^{-t/4}, \\ 0 &= (t-2)e^{-t/4}, \end{aligned}$$

so $t = 2$.

11) d



12) a

$$4\ddot{x} + 8\dot{x} + 13x = 26.$$

The homogeneous equation is $4\ddot{x} + 8\dot{x} + 13x = 0$. The auxiliary equation is $4m^2 + 8m + 13 = 0$:

$$\begin{aligned} 4(m^2 + 2m) + 13 &= 0, \\ 4((m+1)^2 - 1) + 13 &= 0, \\ 4(m+1)^2 - 4 + 13 &= 0, \\ (m+1)^2 &= -\tfrac{9}{4}. \end{aligned}$$

Thus $m = -1 \pm \frac{3}{2}i$, and the complementary function is $x = e^{-t}(A\cos\frac{3}{2}t + B\sin\frac{3}{2}t)$. Try a constant particular integral $x = k$, so $\dot{x} = 0$ and $\ddot{x} = 0$. Substitution gives $13k = 26$, so $k = 2$. Thus the general solution is $x = e^{-t}(A\cos\frac{3}{2}t + B\sin\frac{3}{2}t) + 2$. Using $x = 1$ and $t = 0$ gives $A = -1$. Differentiating,

$$\dot{x} = e^{-t}\left(\tfrac{3}{2}\sin\tfrac{3}{2}t + \tfrac{3}{2}B\cos\tfrac{3}{2}t\right) - e^{-t}\left(-\cos\tfrac{3}{2}t + B\sin\tfrac{3}{2}t\right).$$

Using $\dot{x} = 4$ and $t = 0$ gives $4 = \frac{3}{2}B + 1$, so $B = 2$. The particular solution is

$$x = e^{-t}\left(-\cos\tfrac{3}{2}t + 2\sin\tfrac{3}{2}t\right) + 2.$$

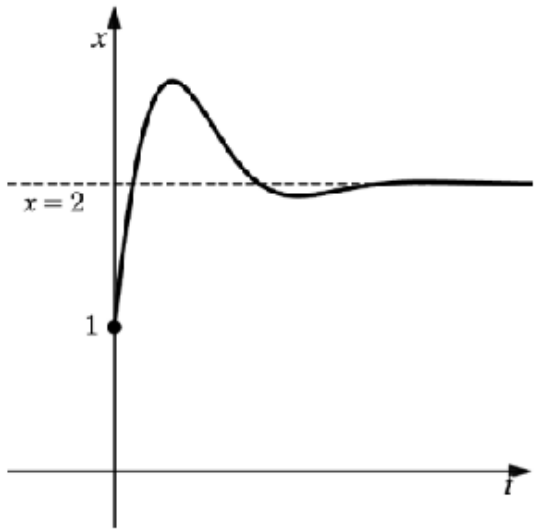
12) b

This is Light Damping (Underdamping) as the auxiliary equation has complex conjugate solutions.

12) c

$\cos\frac{3}{2}t$ and $\sin\frac{3}{2}t$ have period $\frac{2\pi}{3/2}$, so the period of the oscillations is $\frac{4\pi}{3}$.

12) d



13) a

$$\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 17y = 40\cos x.$$

For the homogeneous equation, the auxiliary equation $m^2 + 8m + 17 = 0$ gives $(m+4)^2 = -1$, so $m = -4 \pm i$. The complementary function is $y = e^{-4x}(A\cos x + B\sin x)$. Try $y = p\cos x + q\sin x$, so $y' = -p\sin x + q\cos x$ and $y'' = -p\cos x - q\sin x$. Substituting,

$$\begin{aligned}(-p\cos x - q\sin x) + 8(-p\sin x + q\cos x) + 17(p\cos x + q\sin x) &= 40\cos x,\\ (16p + 8q)\cos x + (-8p + 16q)\sin x &= 40\cos x.\end{aligned}$$

Comparing coefficients gives $16p + 8q = 40$ and $-8p + 16q = 0$. Solving gives $p = 2$ and $q = 1$. The particular integral is $y = 2\cos x + \sin x$. The general solution is

$$y = e^{-4x}(A\cos x + B\sin x) + 2\cos x + \sin x.$$

Using $y = 2$ and $x = 0$ gives $A = 0$, so $y = Be^{-4x}\sin x + 2\cos x + \sin x$. Differentiating,

$$\frac{dy}{dx} = Be^{-4x}\cos x - 4Be^{-4x}\sin x - 2\sin x + \cos x.$$

Using $\frac{dy}{dx} = 2$ and $x = 0$ gives $2 = B + 1$, so $B = 1$. The particular solution is

$$y = e^{-4x}\sin x + 2\cos x + \sin x.$$

13) b

As $x \to \infty$, $e^{-4x} \to 0$, so $y \to 2\cos x + \sin x$, which is simple harmonic form.

14) a

$$4\frac{d^2y}{dx^2} + 9y = \frac{9}{2}x.$$

The homogeneous equation is $4\frac{d^2y}{dx^2} + 9y = 0$. The auxiliary equation is $4m^2 + 9 = 0$, so $m = \pm\frac{3}{2}i$. The complementary function is $y = A\cos\frac{3}{2}x + B\sin\frac{3}{2}x$. Try $y = px + q$, so $y' = p$ and $y'' = 0$. Substitution gives

$$4(0) + 9(px + q) = \frac{9}{2}x \implies (9p)x + (9q) = \frac{9}{2}x.$$

Comparing coefficients, $9p = \frac{9}{2}$, so $p = \frac{1}{2}$, and $9q = 0$, so $q = 0$. The particular integral is $y = \frac{1}{2}x$. Thus the general solution is

$$y = A\cos\frac{3}{2}x + B\sin\frac{3}{2}x + \frac{1}{2}x.$$

Using $y = 4$ and $x = 0$ gives $A = 4$. Differentiating,

$$\frac{dy}{dx} = -6\sin\frac{3}{2}x + \frac{3}{2}B\cos\frac{3}{2}x + \frac{1}{2}.$$

Using $\frac{dy}{dx} = -1$ and $x = 0$ gives $-1 = \frac{3}{2}B + \frac{1}{2}$, so $B = -1$. Therefore the particular solution is

$$y = 4\cos\frac{3}{2}x - \sin\frac{3}{2}x + \frac{1}{2}x.$$

14) b

$$\frac{dy}{dx} = -6\sin\frac{3}{2}x - \frac{3}{2}\cos\frac{3}{2}x + \frac{1}{2}.$$

Stationary points exist when $\frac{dy}{dx} = 0$:

$$\begin{aligned} -6\sin\tfrac{3}{2}x - \tfrac{3}{2}\cos\tfrac{3}{2}x + \tfrac{1}{2} &= 0, \\ 12\sin\tfrac{3}{2}x + 3\cos\tfrac{3}{2}x - 1 &= 0, \\ 3\sqrt{17}\sin\left(\tfrac{3}{2}x + \arctan\tfrac{1}{4}\right) &= 1, \\ \sin\left(\tfrac{3}{2}x + \arctan\tfrac{1}{4}\right) &= \frac{1}{3\sqrt{17}}. \end{aligned}$$

This gives

$$x = \frac{2}{3}\left(\arcsin\frac{1}{3\sqrt{17}} - \arctan\frac{1}{4}\right) = -0.109 \text{ (3 SF)}.$$

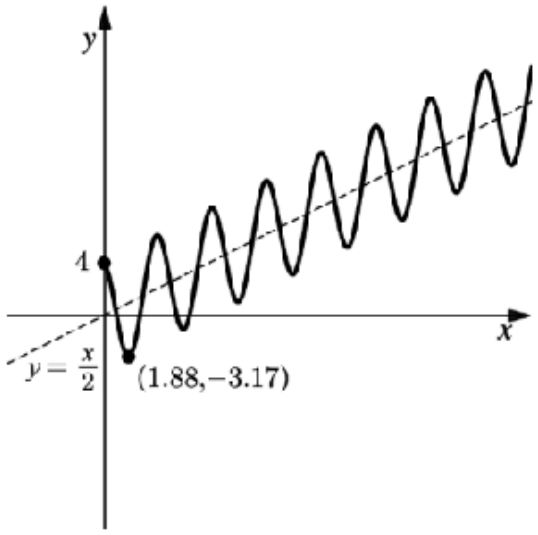
We need $x \geq 0$, so

$$\frac{3}{2}x + \arctan\frac{1}{4} = \pi - \arcsin\frac{1}{3\sqrt{17}},$$

therefore

$$x = \frac{2}{3}\left(\pi - \arcsin\frac{1}{3\sqrt{17}} - \arctan\frac{1}{4}\right) = 1.88 \text{ (3 SF)}.$$

14) c



14) d

As the amplitude is decreasing, the auxiliary equation must have complex roots with negative real part. $4\cos\frac{3}{2}x - \sin\frac{3}{2}x$ has a fixed amplitude and multiplying this by e^{kx} where $k < 0$ ensures that the amplitude of the curve decreases as $x \to \infty$.

14) e

$$y = e^{kx}\left(4\cos\frac{3}{2}x - \sin\frac{3}{2}x\right) + \frac{1}{2}x.$$

Differentiating,

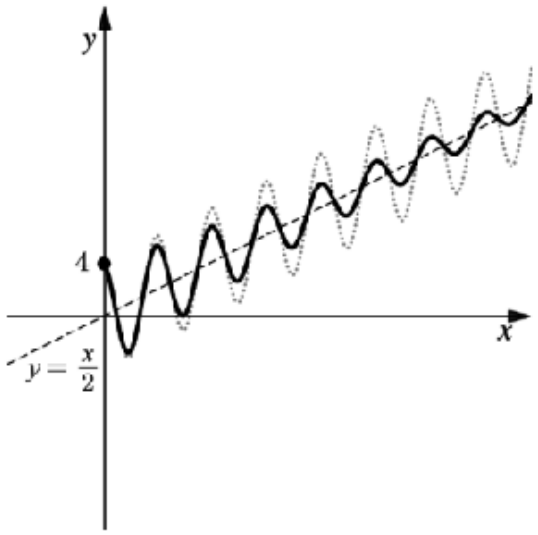
$$\frac{dy}{dx} = e^{kx}\left(-6\sin\frac{3}{2}x - \frac{3}{2}\cos\frac{3}{2}x\right) + ke^{kx}\left(4\cos\frac{3}{2}x - \sin\frac{3}{2}x\right) + \frac{1}{2}.$$

At $x = 0$,

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{x=0} &= e^0\left(-6\sin 0 - \tfrac{3}{2}\cos 0\right) + ke^0(4\cos 0 - \sin 0) + \tfrac{1}{2} \\ &= (0 - \tfrac{3}{2}) + k(4 - 0) + \tfrac{1}{2} \\ &= -\tfrac{3}{2} + 4k + \tfrac{1}{2} = 4k - 1.\end{aligned}$$

If $4k - 1 = -1$, then $k = 0$. As $k < 0$, we cannot have $\left.\frac{dy}{dx}\right|_{x=0} = -1$.

14) f


y
1
x
y = x/2